

NMBS-C7447 at $z = 1.80$ in CPTG

Native-Phase Placement, Local Structural Invariance, and Structural Mode N

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Abstract

NMBS-C7447 is a massive compact quiescent galaxy at redshift $z = 1.80$ whose high stellar mass, small physical size, and large stellar velocity dispersion make it a strong test of how an early bound system should be related to cosmological geometry. The source study reports a stellar mass of order $1.5 \times 10^{11} M_{\odot}$, circularized effective radius $R_e = 1.64 \pm 0.15$ kpc, and stellar velocity dispersion $\sigma_{\star} = 294 \pm 51$ km s $^{-1}$. This paper develops a Curvature Polarization Transport Gravity (CPTG) reduction in which redshift selects the native cosmological phase while the bound-system structural acceleration remains locally determined by the galaxy’s own state. In the declared one-radius pressure-supported reduction, the observations give $a_{\star} \simeq a_{\sigma} = (1.708 \pm 0.613) \times 10^{-9}$ m s $^{-2}$. Independently, the native π branch gives $a_{\pi}(z = 1.80) = 1.64863 \times 10^{-9}$ m s $^{-2}$, so the descriptive cross-scale coordinate is $\Xi_{\star\pi} = a_{\star}/a_{\pi} = 1.036 \pm 0.372$. No preferred value of $\Xi_{\star\pi}$ is imposed. For a fixed intrinsic structural state $\mathcal{S}$, the local-invariance hypothesis is $(\partial a_{\star}/\partial N_{\pi})_{\mathcal{S}} = 0$. This yields an exact phase response $d \ln \Xi_{\star\pi}/d N_{\pi} = 1.44864$ at the observed phase: $a_{\star}$ and R_c remain fixed while a_{π} , R_c , and $\Xi_{\star\pi}$ change with native phase. The measured Sérsic structure and flattening then give a post-solution Structural Mode $N = 3.06465$. Under pure fixed-state phase relocation, the local field and N remain unchanged. The resulting construction separates cosmological placement from bound-system organization and defines a prospective population test of whether $a_{\star}$ retains any residual dependence on native phase after structural state is controlled.

1 NMBS-C7447 as a Cosmological and Dynamical Test Case

NMBS-C7447 is a compact, massive quiescent galaxy at $z = 1.80$ whose combination of stellar mass, size, and velocity dispersion makes it an unusually informative link between early galaxy structure and cosmological interpretation. Its stars are already confined to a remarkably small volume, while its spectrum shows little or no ongoing star formation. The original VLT X-Shooter and HST-WFC3 analysis found a stellar mass of order $1.5 \times 10^{11} M_{\odot}$, a young best-fit stellar population age of about 0.40 Gyr, a circularized effective radius of only 1.64 ± 0.15 kpc, and an aperture-corrected stellar velocity dispersion of 294 ± 51 km s $^{-1}$ [1]. By the observed epoch, a very large stellar mass had therefore already assembled into a dynamically hot, compact system and largely ceased forming new stars.

That combination made NMBS-C7447 important well before CPTG was applied to it. The source study showed that, at fixed dynamical mass, the galaxy is about 2.5 times smaller than comparable low-redshift systems and has a stellar velocity dispersion about 1.8 times higher [1]. Those measurements helped establish that the compactness of high-redshift quiescent galaxies was not merely a photometric artifact: their stellar dynamics independently confirmed that these systems were genuinely denser and structurally different from their present-day counterparts.

For cosmology and galaxy formation, the importance of NMBS-C7447 lies in the combination of early epoch and mature internal structure. A successful framework must accommodate both the existence of such a compact quenched system and the relation between its bound gravitational state and the cosmological geometry in which it is observed. In the standard Λ CDM picture, the redshift is placed on a Friedmann expansion history and galaxy assembly is interpreted within hierarchical structure growth in dark-matter halos; the source study places the compactness and subsequent size evolution of systems such as NMBS-C7447 in that broader hierarchical context [2, 1]. For this object, however, van de Sande et al. do not obtain the quoted dynamical mass from a fitted dark-halo profile: it is a Sérsic-dependent virial estimate constructed from the measured dispersion and radius. CPTG uses the same observations to ask a different question about the division between cosmological phase and bound-system structure.

1.1 Observational Leverage for the Present Test

NMBS-C7447 is particularly useful because the observations constrain cosmological placement and bound-system structure through distinct measurements. Its redshift tells us when the light was emitted. Its stellar velocity dispersion and effective radius tell us how strongly the stellar body is gravitationally supported. Its Sérsic profile and projected axis ratio describe how the stellar source is distributed in radius and shape. These measurements allow the cosmological and local parts of the CPTG calculation to be constructed separately before any cross-scale coordinate is formed.

The source measurements used here are

$$\sigma_{\star} = 294 \pm 51 \text{ km s}^{-1}, \quad R_e = 1.64 \pm 0.15 \text{ kpc}, \quad (1)$$

with Sérsic index $n = 5.3 \pm 0.4$, projected axis ratio $b/a = 0.71 \pm 0.01$, and stellar mass of order $1.5 \times 10^{11} M_{\odot}$ [1]. The published dynamical mass, $(1.7 \pm 0.5) \times 10^{11} M_{\odot}$, is useful as a source-level consistency check, but it is derived from the same dispersion, radius, and Sérsic-dependent virial coefficient and is therefore not an independent CPTG validation target.

2 CPTG Quantities and Test Architecture

The analysis uses several CPTG quantities that must remain conceptually distinct: the object-dependent structural acceleration $a_{\star}$, the native cosmological acceleration a_{π} , the cosmological phase N_{π} , and the post-solution Structural Mode N . The purpose of the present reduction is to ask whether the bound-system quantities can remain functions

of the galaxy’s own structural state while cosmological phase supplies only the external location of that state on the native history.

2.1 Object-Dependent Structural Acceleration $a_\star$

In CPTG, $a_\star$ is the structural acceleration scale associated with a gravitating object. It is not a universal acceleration constant shared by all galaxies, and it is not simply the acceleration at one arbitrary radius. It is the object-dependent scale that enters the curvature-polarization/transport response of the solved system [3]. Different observing configurations can estimate the same physical quantity in different ways. For a pressure-dominated galaxy with only an integrated velocity dispersion and an effective radius, the declared one-radius estimator used here is

$$a_\sigma = \frac{\sigma_\star^2}{R_e}. \quad (2)$$

This is an effective-radius support coordinate. In the present reduction it supplies the one-radius estimate of $a_\star$; it does not redefine $a_\star$ universally. The same structural scale has the CPTG curvature-radius representation

$$R_c = \frac{48\pi c^2}{a_\star}, \quad (3)$$

where lowercase R_c denotes a bound-system structural curvature scale, not the physical radius of the visible galaxy.

2.2 Native Cosmological Phase and Acceleration

CPTG has one native geometric π branch that supplies cosmological phase. Its phase coordinate is

$$N_\pi = \ln a, \quad (4)$$

and its acceleration coordinate is $a_\pi(a)$. The associated uppercase curvature-transport radius is

$$R_C(a) = \frac{48\pi c^2}{a_\pi(a)}. \quad (5)$$

Thus $a_\star$ characterizes the bound galaxy, whereas a_π characterizes native cosmological geometry. Redshift fixes N_π and a_π without assigning the galaxy’s $a_\star$.

2.3 Local Structural Invariance at Fixed Intrinsic State

Let $\mathcal{S}$ denote the intrinsic bound-system state held fixed in a comparison: its baryonic source distribution, geometry, and support state. The symbol is a bookkeeping device, not an additional fit parameter. The local structural-invariance hypothesis investigated here is

$$a_\star = a_\star(\mathcal{S}), \quad \left. \frac{\partial a_\star}{\partial N_\pi} \right|_{\mathcal{S}} = 0. \quad (6)$$

The statement is deliberately conditional. It does not claim that a real galaxy is frozen through cosmic history. If the galaxy itself evolves, then $\mathcal{S}$ changes and $a_\star$ may change with it. The total derivative can be written schematically as

$$\frac{da_\star}{dN_\pi} = \left. \frac{\partial a_\star}{\partial N_\pi} \right|_{\mathcal{S}} + \sum_j \frac{\partial a_\star}{\partial \mathcal{S}_j} \frac{d\mathcal{S}_j}{dN_\pi}. \quad (7)$$

Under Eq. (6), only genuine structural evolution contributes to the second term. Pure relocation of the same state on the native phase coordinate does not rescale $a_\star$ by itself.

Equation (6) immediately implies

$$\left. \frac{\partial R_c}{\partial N_\pi} \right|_{\mathcal{S}} = 0, \quad \left. \frac{\partial H_{\text{struct}}}{\partial N_\pi} \right|_{\mathcal{S}} = 0, \quad (8)$$

with

$$H_{\text{struct}} = \frac{16\pi}{3} \frac{a_\star}{c}. \quad (9)$$

The bound-system amplitude remains local; the cosmological coordinates continue to evolve on their own branch.

2.4 Cross-Scale Phase Coordinate $\Xi_{\star\pi}$

The common curvature-radius form allows a dimensionless comparison to be constructed after the two scales have been obtained:

$$\Xi_{\star\pi}(a) \equiv \frac{R_C(a)}{R_c} = \frac{a_\star}{a_\pi(a)}. \quad (10)$$

In this paper $\Xi_{\star\pi}$ is a descriptive phase coordinate. No preferred value is imposed. It records where the locally determined bound-system scale lies relative to the native cosmological scale at a chosen phase. Its phase dependence will become the principal derived consequence of Eq. (6).

2.5 Post-Solution Structural Mode N

The cross-scale coordinate does not describe the internal organization of the solved galaxy. Structural Mode N provides that downstream information. Structural Mode is evaluated only after the baryonic source, $a_\star$, and nonlinear CPTG field have been fixed. It measures where radial curvature-polarization/transport structure changes most strongly. Lower N corresponds to a response whose dominant structural variation is weighted farther outward; higher N corresponds to a response organized progressively farther inward [5]. It is therefore a structural diagnostic, not another free fitting parameter.

The two N symbols answer different questions:

$$N_\pi : \text{ when the galaxy is represented,} \quad N : \text{ where the solved field is organized.} \quad (11)$$

2.6 Separation of Observables and Derived Coordinates

The reduction preserves the sequence

$$\begin{aligned}
&\text{redshift} \longrightarrow N_\pi, a_\pi, R_C, \\
&\text{stellar support} \longrightarrow a_\star, R_c, \\
&\text{baryonic structure} \longrightarrow g(r) \longrightarrow N, \\
&\text{only afterward: } \Xi_{\star\pi} = \frac{a_\star}{a_\pi}.
\end{aligned} \tag{12}$$

The cross-scale ratio is therefore not used to determine either side of the calculation. For orientation, the observed galaxy gives

$$a_\star \simeq (1.70805 \pm 0.61284) \times 10^{-9} \text{ m s}^{-2}, \tag{13}$$

$$a_\pi(z = 1.80) = 1.64863 \times 10^{-9} \text{ m s}^{-2}. \tag{14}$$

Their central values happen to differ by 3.60%, but the observational uncertainty on the local estimate is about 36%. That numerical proximity is reported as a property of the observed state, not as a benchmark imposed by the present formulation.

3 Observational Inputs and Reduction Strategy

Table 1 collects the observational quantities that enter the analysis and distinguishes them from quantities derived within CPTG. The primary calculation retains the published physical quantities under the source paper’s original distance convention; a later section tests how much the local structural estimate moves if those distance-dependent quantities are re-expressed through current CPTG comparison coordinates.

Quantity	Adopted value	Role in this paper
Spectroscopic redshift	$z = 1.80$	Selects the native CPTG phase
Aperture-corrected stellar dispersion	$294 \pm 51 \text{ km s}^{-1}$	Pressure-support measurement
Circularized F160W effective radius	$1.64 \pm 0.15 \text{ kpc}$	One-radius support scale; area-equivalent Sérsic baseline
Sérsic index	$n = 5.3 \pm 0.4$	Radial stellar-source shape
Projected axis ratio	$q_{\text{proj}} = b/a = 0.71 \pm 0.01$	Oblate source-geometry constraint
Stellar mass	$M_\star \simeq 1.5 \times 10^{11} M_\odot$	Radial source normalization; source paper adopts ~ 0.2 dex systematic uncertainty
Published dynamical mass	$(1.7 \pm 0.5) \times 10^{11} M_\odot$	Shared-input virial consistency reference, not an independent validation observable

Table 1: Observed NMBS-C7447 quantities used in the local structural and Structural Mode calculations [1].

The reduction has five stages:

Stage	Observational information	CPTG output	Physical question
1	Redshift z	$a, N_\pi, E_\pi, a_\pi, R_C$	Where is the observation on native cosmological geometry?
2	$\sigma_\star$ and R_e	$a_\sigma \rightarrow a_\star, R_c$	What local structural acceleration is indicated by stellar support?
3	Fixed structural state $\mathcal{S}$	phase law for $\Xi_{\star\pi}$	What changes if native phase changes but the bound state does not?
4	$M_\star, n, q_{\text{proj}}$	$g(r), \lambda, N$	Where is the solved local field organized?
5	Phase sweep at fixed $\mathcal{S}$	invariant $a_\star, R_c, g(r), N$	Does cosmological phase alone alter the bound solution?

Table 2: Inference sequence for the NMBS-C7447 local-structural-invariance test.

4 Native-Phase Placement of NMBS-C7447

The cosmological part of the calculation begins with the measured redshift alone. Galaxy mass, radius, dispersion, Sérsic structure, and Structural Mode do not enter this stage. The complementary native weights are [3, 4]

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}, \quad \mathcal{B}_\pi + \mathcal{T}_\pi = 1, \quad (15)$$

with

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}. \quad (16)$$

The observed redshift selects

$$a = \frac{1}{1+z} = 0.3571428571, \quad N_\pi = \ln a = -1.029619417, \quad (17)$$

and

$$E_\pi(z = 1.80) = 2.960729196. \quad (18)$$

The native present acceleration anchor is

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} = 5.568327997 \times 10^{-10} \text{ m s}^{-2}, \quad (19)$$

so

$$a_\pi(z) = a_{\pi,0} E_\pi(z) = 1.648631127 \times 10^{-9} \text{ m s}^{-2}. \quad (20)$$

The corresponding native curvature-transport radius is

$$R_C(z) = \frac{48\pi c^2}{a_\pi(z)} = 266.4149 \text{ Gpc}. \quad (21)$$

The native rate is

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a), \quad H_0^{(\pi)} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (22)$$

with equivalent acceleration form

$$H_\pi(a) = \frac{a_\pi(a)}{c\sqrt{\mathcal{B}_\pi}}. \quad (23)$$

Direct integration of $dt = da/[aH_\pi(a)]$ gives

$$t_\pi(a) = \frac{2}{\pi H_0^{(\pi)} \sqrt{\mathcal{B}_\pi}} \sinh^{-1} \left[\sqrt{\frac{\mathcal{B}_\pi}{\mathcal{T}_\pi}} a^{\pi/2} \right], \quad (24)$$

and hence

$$t_\pi(z = 1.80) = 3.11117 \text{ Gyr}, \quad t_{\pi,0} = 12.75412 \text{ Gyr}. \quad (25)$$

These quantities locate the observation on the native history. They do not determine the galaxy's local $a_\star$. The source redshift is retained at its published value $z = 1.80$ because no redshift uncertainty is used in the source reduction; the local observational uncertainty budget is dominated by $\sigma_\star$ and R_e .

5 Pressure-Supported Structural Acceleration

Once the native phase has been located, the analysis turns to the galaxy itself. In the present pressure-dominated one-radius reduction,

$$a_\sigma = \frac{\sigma_\star^2}{R_e}. \quad (26)$$

Dimensionally this is an acceleration; physically it is a compact summary of the stellar support required at the effective-radius scale. CPTG does not identify a_σ with $a_\star$ by definition in every system. Here it is explicitly adopted as the one-radius estimator of the object's structural acceleration scale. The available integrated kinematics do not separately resolve ordered rotation, inclination, or orbital anisotropy, so those support-decomposition uncertainties remain outside the scalar estimator. A future resolved Jeans analysis could replace this estimate without changing the meaning of $a_\star$.

Using the published midpoint values gives

$$a_\star \simeq a_\sigma = \frac{(294 \text{ km s}^{-1})^2}{1.64 \text{ kpc}} = 1.708048772 \times 10^{-9} \text{ m s}^{-2}. \quad (27)$$

For independent quoted errors in $\sigma_\star$ and R_e ,

$$\left(\frac{\delta a_\sigma}{a_\sigma} \right)^2 = \left(2 \frac{\delta \sigma_\star}{\sigma_\star} \right)^2 + \left(\frac{\delta R_e}{R_e} \right)^2, \quad (28)$$

which gives

$$a_\star \simeq a_\sigma = (1.708 \pm 0.613) \times 10^{-9} \text{ m s}^{-2}. \quad (29)$$

The fractional observational scale is about 36%, dominated by the dispersion uncertainty.

The associated local structural curvature scale is

$$R_c = \frac{48\pi c^2}{a_\star} = 257.1471 \text{ Gpc}, \quad (30)$$

with linearized observational scale

$$R_c = 257.15 \pm 92.26 \text{ Gpc}. \quad (31)$$

This is a CPTG curvature-transport scale, not the physical size of the stellar galaxy.

6 Local Structural Invariance and the Native-Phase Response

The local-invariance hypothesis now has a concrete observed anchor. At the measured NMBS-C7447 state, a_* and R_c are fixed by the bound-system reduction above. If that same structural state $\mathcal{S}$ is represented at another native phase without changing the galaxy itself, Eq. (6) leaves those local quantities unchanged while the native cosmological quantities continue to evolve.

Because

$$a_\pi(a) = a_{\pi,0} E_\pi(a), \quad (32)$$

Eq. (10) becomes

$$\Xi_{*\pi}(a) = \frac{a_*}{a_{\pi,0}} \frac{1}{E_\pi(a)}. \quad (33)$$

At the observed phase,

$$\Xi_{*\pi}(z = 1.80) = \frac{1.708048772 \times 10^{-9}}{1.648631127 \times 10^{-9}} = 1.036040594. \quad (34)$$

Propagating the same σ_* and R_e uncertainties gives

$$\Xi_{*\pi}(z = 1.80) = 1.036 \pm 0.372. \quad (35)$$

The central value is 3.60% above unity, but the uncertainty is more than ten times larger than that offset. No equality condition is used in this paper; Eq. (35) is simply the observed cross-scale location of the galaxy.

6.1 Exact Native-Phase Derivative

Since $N_\pi = \ln a$, differentiating Eq. (33) gives

$$\left. \frac{d \ln \Xi_{*\pi}}{d N_\pi} \right|_{\mathcal{S}} = - \frac{d \ln E_\pi}{d N_\pi}. \quad (36)$$

Using Eq. (16),

$$\left. \frac{d \ln \Xi_{*\pi}}{d N_\pi} \right|_{\mathcal{S}} = \frac{\pi}{2} \frac{\mathcal{T}_\pi a^{-\pi}}{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}}. \quad (37)$$

At $z = 1.80$,

$$\left. \frac{d \ln \Xi_{*\pi}}{d N_\pi} \right|_{\mathcal{S}, z=1.80} = 1.448641818. \quad (38)$$

This derivative is determined entirely by the native branch once the local state is held fixed. It approaches $\pi/2$ toward very early native phase and equals $1/2$ at the normalized present phase $a = 1$. The cross-scale coordinate therefore evolves even though the local structural acceleration does not.

The same result can be anchored directly to the observed state:

$$\Xi_{*\pi}(a) = \Xi_{*\pi}(a_{\text{obs}}) \frac{E_\pi(a_{\text{obs}})}{E_\pi(a)}. \quad (39)$$

At $a = 1$ this fixed-state comparison gives

$$\Xi_{\star\pi}(a = 1) = 3.0674 \pm 1.1006. \quad (40)$$

This is a phase-coordinate comparison for the same structural state, not a prediction that the actual galaxy will retain its present mass profile and support indefinitely.

6.2 Dependence on the Source Distance Convention

The published radius and stellar mass were converted to physical units using the source paper’s flat- Λ CDM convention, $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$ [1]. The primary result preserves those published quantities. A coordinate-sensitivity calculation then re-expresses the distance-dependent quantities through the current CPTG acoustic and luminosity comparison coordinates.

For any comparison branch b ,

$$D_{A,b}(z) = \frac{c}{(1+z)H_{0,b}} \int_0^z \frac{dz'}{E_b(z')}. \quad (41)$$

The current acoustic comparison coordinate uses

$$p_{\text{ac}} = 3 - \frac{\pi}{100}, \quad E_{\text{ac}}^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad H_{0,\text{ac}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (42)$$

while the luminosity comparison coordinate uses

$$\Omega_{m,\text{ac}} = \frac{p_{\text{ac}}}{3\pi}, \quad \Omega_{\Lambda,\text{ac}} = 1 - \Omega_{m,\text{ac}}, \quad E_{\text{lum}}^2 = \Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}, \quad (43)$$

with $H_{0,\text{lum}} = 73.17 \text{ km s}^{-1} \text{ Mpc}^{-1}$. These are comparison coordinates evaluated at the same native phase, not additional native cosmologies.

At $z = 1.80$,

$$D_A^{\Lambda 70} = 1741.846 \text{ Mpc}, \quad (44)$$

$$D_{A,\text{ac}} = 1791.516 \text{ Mpc}, \quad (45)$$

$$D_{A,\text{lum}} = 1656.481 \text{ Mpc}, \quad (46)$$

$$D_{A,\text{mid}} = 1723.998 \text{ Mpc}. \quad (47)$$

Re-expressing R_e by $s_D = D_A/D_A^{\Lambda 70}$ and the constant- M/L stellar normalization by s_D^2 gives

Distance coordinate	R_e (kpc)	$a_\star$ (10^{-9} m s^{-2})	$\Xi_{\star\pi}$
Published source	1.64000	1.7081 ± 0.6128	1.0360 ± 0.3717
Acoustic endpoint	1.68677	1.6607 ± 0.5958	1.0073 ± 0.3614
Symmetric midpoint	1.62320	1.7257 ± 0.6192	1.0468 ± 0.3756
Luminosity endpoint	1.55963	1.7961 ± 0.6444	1.0894 ± 0.3909

Table 3: Distance-coordinate sensitivity of the local structural estimate and its descriptive native-phase ratio. The quoted uncertainties propagate the observed $\sigma_\star$ and R_e errors at fixed comparison coordinate; the branch span itself is deterministic.

The local acceleration changes by only a few percent across the full present acoustic-to-luminosity span, much less than the current observational uncertainty.

7 Fixed-State Native-Phase Sweep

The defining thought experiment of the present paper changes only native phase. The measured bound state $\mathcal{S}$ is held fixed. Consequently

$$a_\star = (1.708 \pm 0.613) \times 10^{-9} \text{ m s}^{-2}, \quad R_c = 257.15 \pm 92.26 \text{ Gpc} \quad (48)$$

remain the same throughout the sweep, while E_π , a_π , R_C , and $\Xi_{\star\pi}$ change according to the native branch.

z	N_π	E_π	R_C (Gpc)	$\Xi_{\star\pi}$
0.0	0.0000	1.0000	788.782	3.0674 ± 1.1006
0.5	-0.4055	1.3489	584.770	2.2741 ± 0.8159
1.0	-0.6931	1.8684	422.179	1.6418 ± 0.5891
1.8	-1.0296	2.9607	266.415	1.0360 ± 0.3717
2.0	-1.0986	3.2745	240.884	0.9368 ± 0.3361
3.0	-1.3863	5.0470	156.289	0.6078 ± 0.2181
4.0	-1.6094	7.1172	110.828	0.4310 ± 0.1546

Table 4: Pure native-phase coordinate sweep of the same NMBS-C7447 structural state. The local $a_\star$ and R_c are held fixed; only the native cosmological coordinates vary. The $z = 1.80$ row is the actual observed phase. The other rows are fixed-state comparison coordinates, not a reconstruction of the galaxy’s physical evolution.

The table makes the separation explicit. At the actual observed phase, the local and native acceleration coordinates happen to be numerically close. That proximity is not preserved by Eq. (6): as native phase changes while the galaxy is held fixed, the cosmological scale moves past the unchanged local scale and $\Xi_{\star\pi}$ changes accordingly.

7.1 Local Field Invariance Under Pure Phase Relocation

The current radial CPTG galaxy equation contains the local baryonic source, $a_\star$, and the derived local R_c ; it contains no explicit N_π once those local quantities have been fixed. Therefore a pure phase relocation of the same $\mathcal{S}$ gives

$$\left. \frac{\partial g(r)}{\partial N_\pi} \right|_{\mathcal{S}} = 0. \quad (49)$$

Any post-solution functional constructed solely from that unchanged local field is also invariant. In particular,

$$\left. \frac{\partial \lambda}{\partial N_\pi} \right|_{\mathcal{S}} = 0, \quad \left. \frac{\partial N}{\partial N_\pi} \right|_{\mathcal{S}} = 0. \quad (50)$$

The numerical evidence package evaluates the same retained high-resolution local solution across the phase labels of Table 4; the Structural Mode spread is exactly zero to the stored numerical precision. This is a consistency check of the reduction, not evidence that an actual galaxy cannot evolve.

8 Radial Field Reconstruction and Structural Mode

The local structural acceleration is only one number. The measured Sérsic structure allows the analysis to ask a second question: how is the resulting CPTG field organized across the galaxy? Structural Mode N provides that downstream description [5].

For a solved radial acceleration field $g(r)$, define

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad \kappa_{\text{struct}} = 1 \text{ kpc}, \quad (51)$$

and

$$w(r) = \left| \frac{dC_g}{dr} \right|. \quad (52)$$

The organization length and Structural Mode are

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}, \quad N = \frac{R}{\lambda}. \quad (53)$$

Larger N therefore places the retained structural change farther inward as a fraction of the registered outer radius.

8.1 Stellar-Source Reconstruction from the Observed Sérsic Profile

The source study provides the Sérsic index and projected axis ratio needed to extend the one-radius calculation into a radial reconstruction. The calculation assumes that stellar mass follows the fitted F160W Sérsic light profile with spatially constant stellar mass-to-light ratio. The source analysis does not provide a measured radial gas-mass profile for NMBS-C7447. The present radial CPTG source is therefore treated as stellar dominated and normalized to the reported stellar mass M_* ; any unmeasured non-stellar baryonic component lies outside this Structural Mode reconstruction.

For the observed profile,

$$S(s) = \exp \left[-b_n \left(\left(\frac{s}{R_e} \right)^{1/n} - 1 \right) \right], \quad (54)$$

with $n = 5.3$. Because an ideal Sérsic model has no finite edge, the Structural Mode readout is registered at $R \equiv R_{95}$, where

$$\frac{\gamma(2n, b_n(R_{95}/R_e)^{1/n})}{\Gamma(2n)} = 0.95, \quad \frac{\gamma(2n, b_n)}{\Gamma(2n)} = \frac{1}{2}. \quad (55)$$

This is a model registration, not a claim that stellar mass is directly resolved to 20–24 kpc. The area-equivalent spherical reconstruction gives

$$R_{95, \text{circ}} = 20.00054 \text{ kpc}. \quad (56)$$

For the axisymmetric reconstruction,

$$R_{e,\text{maj}} = \frac{R_{e,\text{circ}}}{\sqrt{q_{\text{proj}}}} = 1.94632 \text{ kpc}. \quad (57)$$

For similar oblate stellar-density surfaces,

$$m^2 = R^2 + \frac{z^2}{q_{\text{int}}^2}, \quad q_{\text{proj}}^2 = \cos^2 i + q_{\text{int}}^2 \sin^2 i, \quad (58)$$

so $q_{\text{proj}} = 0.71$ admits $0 < q_{\text{int}} \leq 0.71$, with $q_{\text{int}} = 0.71$ at the edge-on endpoint.

Let

$$L_S = 2\pi \int_0^\infty S(s) s \, ds. \quad (59)$$

The oblate density is

$$\rho_q(m) = -\frac{M_\star}{\pi q_{\text{int}} L_S} \int_m^\infty \frac{dS/ds}{\sqrt{s^2 - m^2}} ds, \quad (60)$$

and the exact equatorial source field of nested similar oblate homoeoids is

$$g_{\text{bar,obl}}(R) = \frac{4\pi G q_{\text{int}}}{R} \int_0^R \frac{\rho_q(m) m^2 \, dm}{\sqrt{R^2 - (1 - q_{\text{int}}^2) m^2}}. \quad (61)$$

The $q_{\text{int}} = 1$ limit reduces to $GM_\star(< R)/R^2$.

At every radius, the same observed local structural estimate from Eq. (27) enters the registered CPTG response law,

$$g = g_{\text{bar}} + \frac{\sqrt{a_\star g_{\text{bar}}} + g_{\text{geom}}}{[1 + (g/a_\star)^{123/50}]^{11/50}}, \quad (62)$$

with

$$g_{\text{geom}}(r) = \frac{1}{2} a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad R_c = \frac{48\pi c^2}{a_\star}. \quad (63)$$

No Hubble type, CSMI class, cross-scale phase coordinate, or native phase variable is inserted into this radial solve. The field is determined from the local source and local structural scale.

8.2 Structural Mode Result and Geometry Dependence

The simplest area-equivalent spherical baseline gives

$$\lambda_{\text{sph,circ}} = 6.51415 \text{ kpc}, \quad N_{\text{sph,circ}} = 3.07032. \quad (64)$$

A spherical control using the same semimajor-axis Sérsic scale gives

$$R_{95,\text{maj}} = 23.73628 \text{ kpc}, \quad N_{\text{sph,maj}} = 3.04944. \quad (65)$$

The measured-flattening edge-on solution gives

$$\lambda_{\text{obl}} = 7.74518 \text{ kpc}, \quad N_{\text{obl}} = 3.06465. \quad (66)$$

The explicitly calculated thin-geometry stress point $q_{\text{int}} = 0.005$ gives

$$N = 3.11552, \quad (67)$$

and continuing to $q_{\text{int}} = 2 \times 10^{-4}$ gives

$$N = 3.11592, \quad (68)$$

which is a numerical near-thin point rather than an analytic $q_{\text{int}} = 0$ solution.

Structural reconstruction	N	CSMI readout
Area-equivalent spherical baseline	3.07032	Early Spiral
Same-semimajor spherical geometry control	3.04944	Intermediate Spiral
Measured-flattening oblate edge-on endpoint	3.06465	Early Spiral
Thin-geometry stress point, $q_{\text{int}} = 0.005$	3.11552	Early Spiral
Near-thin numerical point, $q_{\text{int}} = 2 \times 10^{-4}$	3.11592	Early Spiral

Table 5: Post-solution Structural Mode results. CSMI is a downstream catalog lookup applied only after continuous N has been obtained.

The CSMI labels are descriptive interval names, not morphology inputs. The measured Sérsic profile and axis ratio are quantitative source observables; the categorical class is assigned only after N is returned by the solved field.

The outer-light registration is also tested directly:

$$N(R_{90}) = 3.08076, \quad N(R_{95}) = 3.06465, \quad N(R_{99}) = 3.08343. \quad (69)$$

The continuous mode therefore remains near $N \simeq 3.07$ over a broad 90–99 percent Sérsic-light registration range.

9 Uncertainty Structure, Sensitivity, and Falsifiability

The local acceleration and cross-scale coordinate are presently measurement limited. Because $a_\sigma \propto \sigma_\star^2$, the $\pm 51 \text{ km s}^{-1}$ dispersion uncertainty dominates both $a_\star$ and $\Xi_{\star\pi}$. A substantially more precise stellar dispersion would therefore provide the most direct improvement to the local structural test.

Structural Mode is less sensitive to any one quoted source perturbation because it measures radial organization rather than absolute amplitude. For the measured-flattening edge-on endpoint,

Perturbed input	Returned N interval	Comment
$\sigma_\star = 294 \pm 51 \text{ km s}^{-1}$	3.05682–3.07765	Pressure-support estimator changes
$R_e = 1.64 \pm 0.15 \text{ kpc}$	3.05830–3.07275	Radius affects support and source scale
$n = 5.3 \pm 0.4$	3.05769–3.07046	Radial concentration sensitivity
$\log_{10} M_\star \pm 0.2 \text{ dex}$	3.05339–3.08142	Source-normalization sensitivity
$q_{\text{proj}} = 0.71 \pm 0.01$	3.06448–3.06482	Edge-on geometry sensitivity

Table 6: One-at-a-time Structural Mode sensitivities at the measured-flattening edge-on endpoint. These are deterministic sensitivity intervals, not confidence intervals.

A linearized combination of the quoted $\sigma_\star$, R_e , n , and q_{proj} measurement scales gives an illustrative N scale of about 0.0133. Including the source paper’s 0.2 dex stellar-mass systematic as an additional symmetric scale gives about 0.0181. Neither is a formal confidence interval because a joint covariance model is unavailable and the stellar-mass term is systematic.

The deterministic simultaneous signed-corner scan gives

$$3.03026 \leq N_{\text{edge,corner}} \leq 3.13356, \quad (70)$$

and allowing the same source corners to span the edge-on to near-thin geometry family gives

$$3.03026 \lesssim N_{\text{source+geom}} \lesssim 3.21116. \quad (71)$$

These are deliberately conservative deterministic envelopes, not statistical confidence intervals. Continuous N remains well defined, while a hard CSMI label can move when all source and geometry extremes are compounded.

The observed-phase $\Xi_{\star\pi}$ and Structural Mode are also not statistically independent even though neither is used to tune the other. Both inherit $\sigma_\star$ and R_e . Under an illustrative independent-input linearization, including the 0.2 dex mass scale only in the N variance,

$$\rho_{\Xi_{\star\pi},N} \simeq -0.39. \quad (72)$$

This is a diagnostic shared-input covariance, not an inferential population correlation.

9.1 Prospective Test of Local Structural Invariance

A single galaxy observed at one phase cannot establish Eq. (6). The hypothesis becomes testable across a population. Define the fixed-state phase-response coefficient

$$\mathcal{P}_\star \equiv \left. \frac{\partial \ln a_\star}{\partial N_\pi} \right|_S. \quad (73)$$

The present formulation predicts

$$\mathcal{P}_\star = 0. \quad (74)$$

Operationally, future pressure-supported systems should be compared only after structural state is controlled using measured mass profile, size, geometry, support state, and post-solution organization. Any residual systematic dependence of $a_\star$ on N_π after those quantities are accounted for would falsify the strict local-invariance hypothesis or demonstrate that the chosen structural-state description is incomplete.

The cross-scale coordinate itself is not expected to be phase invariant. Once $a_\star$ is locally fixed, Eq. (37) predicts its native-phase evolution exactly. A future sample can therefore test two linked statements simultaneously: whether $a_\star$ has zero residual phase response at fixed structure, and whether the derived $\Xi_{\star\pi}$ then follows the branch response implied by E_π .

10 Physical Interpretation

10.1 The Observed NMBS-C7447 State

The galaxy can be summarized at its observed phase by

$$(\Xi_{\star\pi}, N_{\text{obl}}) = (1.036 \pm 0.372, 3.06465). \quad (75)$$

The first number locates the locally inferred acceleration relative to the native cosmological acceleration at $z = 1.80$; the second describes the radial organization of the solved local field. Their roles remain distinct.

The numerical proximity of $a_{\star}$ and a_{π} at the observed phase is noteworthy but not structurally privileged in this reduction. The fixed-state phase sweep shows why: with the local galaxy held unchanged, $a_{\star}$ stays fixed while a_{π} evolves, so the ratio naturally moves from 1.036 ± 0.372 at $z = 1.80$ to 3.067 ± 1.101 at the normalized present phase. The local structural state does not have to follow the cosmological acceleration coordinate merely because both can be written in the same curvature-radius family.

Structural Mode reinforces the separation. The same local source and local $a_{\star}$ give $N = 3.06465$, and a pure change of native phase cannot alter that number because N_{π} is absent from the fixed local field equation. Actual changes in mass distribution, geometry, or support would require a new solve and could change N ; cosmological relabeling alone does not.

10.2 Relation to Standard Λ CDM Inference

CPTG and Λ CDM begin from the same measured redshift, velocity dispersion, and HST structural quantities. Their distinction lies in how those observations are organized within the gravitational and cosmological inference.

In the standard Λ CDM framework, redshift is placed on a Friedmann cosmological background and galaxy formation is ordinarily interpreted within a dark-matter-halo hierarchy. The compactness of NMBS-C7447 is then part of the broader problem of early assembly, quenching, merger history, and subsequent size growth [2, 1]. For NMBS-C7447 specifically, the source paper does not fit a dark-halo mass profile; its quoted dynamical mass is a Sérsic-dependent virial estimate built from the measured radius and stellar dispersion. This paper does not compute a Λ CDM likelihood comparison or attempt to falsify the standard framework with one galaxy.

CPTG separates native geometry from the bound-system solution. Redshift selects the native π -branch phase. The galaxy's own support supplies the one-radius estimate of its object-dependent $a_{\star}$. The local baryonic source then generates the nonlinear polarization/transport field without inserting a dark-halo gravitational source, and Structural Mode is read only afterward. In the local-invariance formulation, native phase is not an automatic multiplicative rescaling of the bound-system acceleration. That is the principal conceptual distinction examined here.

10.3 Scope and Evidentiary Status

The present paper establishes a mathematically closed single-object construction, not a population confirmation of phase independence. NMBS-C7447 supplies one measured structural state at one observed phase. From that state, the local-invariance assumption produces exact consequences for the fixed-state phase sweep and for the cross-scale coordinate $\Xi_{\star\pi}$, but only a multi-object sample spanning independent native phases can determine whether the assumed absence of residual phase dependence is realized in nature.

The decisive future test is therefore not whether one special ratio is recovered. It is whether structurally comparable bound systems continue to return local $a_\star$ values determined by their own state without an additional systematic N_π term. If a residual phase dependence remains after source structure, support, and Structural Mode are controlled, Eq. (6) fails in its strict form. If no such dependence is required, the separation between native cosmological phase and local bound-system structure gains direct observational support.

11 Conclusion

NMBS-C7447 is an unusually informative galaxy because the observations connect an early cosmological epoch to a directly measured compact stellar structure. The source data show a $\sim 1.5 \times 10^{11} M_\odot$ quiescent galaxy with $R_e = 1.64 \pm 0.15$ kpc and $\sigma_\star = 294 \pm 51$ km s $^{-1}$. Those measurements allow CPTG to separate cosmological placement from local bound-system inference.

Redshift alone fixes the native CPTG phase and gives

$$a_\pi(z = 1.80) = 1.64863 \times 10^{-9} \text{ m s}^{-2}. \quad (76)$$

The galaxy independently supplies the one-radius structural estimate

$$a_\star \simeq a_\sigma = (1.708 \pm 0.613) \times 10^{-9} \text{ m s}^{-2}. \quad (77)$$

The corresponding observed-phase comparison coordinate is

$$\Xi_{\star\pi} = 1.036 \pm 0.372. \quad (78)$$

No preferred value of this ratio is assumed.

The defining local-invariance statement is

$$\left. \frac{\partial a_\star}{\partial N_\pi} \right|_{\mathcal{S}} = 0. \quad (79)$$

At fixed structural state it leads to the exact phase law

$$\Xi_{\star\pi}(a) = \frac{a_\star}{a_{\pi,0} E_\pi(a)}, \quad (80)$$

with

$$\left. \frac{d \ln \Xi_{\star\pi}}{d N_\pi} \right|_{z=1.80, \mathcal{S}} = 1.44864. \quad (81)$$

Thus the native cosmological coordinates change while the bound $a_\star$ and R_c remain fixed under pure phase relocation.

The radial reconstruction gives

$$N_{\text{obl}} = 3.06465, \tag{82}$$

with the same explicit source and geometry sensitivity controls used throughout the calculation. Because the local field equation contains no explicit N_π once the local state is fixed, that Structural Mode is unchanged by the pure phase sweep as well.

The broader significance lies in the separation of roles. CPTG can place a galaxy on native cosmological geometry without using that cosmological phase to redefine the galaxy’s local structural amplitude. The future empirical question is whether this separation survives across structurally controlled systems spanning native phase. That population test will determine whether local structural invariance is a genuine property of bound CPTG systems or whether an additional phase dependence is required by the observations.

Data, Code, and Evidence Availability

The observational authority for NMBS-C7447 is van de Sande et al. [1]. Standard Λ CDM language is used only for contextual contrast and for documenting the source paper’s distance convention; the governing CPTG cosmological construction is the native π branch. The native branch, structural-acceleration hierarchy, and local/native curvature-radius definitions follow the current unified framework and π -branch comparison-coordinate paper [3, 4]. Structural Mode and the downstream CSMI lookup follow the current Structural Mode validation [5].

The exact numerical evidence authority for the native-phase calculation, pressure-support propagation, local structural scale, fixed-state phase law, $\Xi_{\star\pi}$ sweep, distance-coordinate sensitivity, Sérsic and oblate source reconstruction, nonlinear field solve, Structural Mode evaluation, source and geometry sensitivity scans, shared-input covariance, monotonic- C_g verification, and direct $|dC_g/dr|$ cross-check is archived as one self-contained evidence package. The archive contains the authoritative script and output, SHA-256 manifest, reproduction instructions, execution-environment record, and clean rerun verification.

CPTG_NMBS-C7447_LocalStructuralInvariance_Evidence_20260904_r01_FINAL_EVIDENCE.zip

SHA-256: 875c079e44e6ce4e5963987591bd7dfc8e4b7d36ecee046ab45b743702b44a2

The public CPTG source repository is available through the supporting resource [6].

11.1 Companion Mathematical Comparison

A comparison experiment was also carried out with this NMBS-C7447 analysis:

NMBS-C7447 at $z = 1.80$ in CPTG: Native-Phase Placement, Cross-Scale Locking, and Structural Mode N .

Because both papers begin from the same observed NMBS-C7447 measurements and return the same observed-state CPTG quantities, the comparison used that common solution as the starting point and examined only the different mathematical treatment of the galaxy away from its measured native phase. The experiment compared the behavior of $a_\star$, R_c , the structural-to-native acceleration ratio $a_\star/a_\pi$, the reconstructed CPTG radial field, and Structural Mode N under the two prescriptions.

The purpose was to determine whether native locking and local structural invariance are mathematically equivalent descriptions or genuinely different native-phase trajectories, and to identify precisely where their predictions separate without altering the definitions used in either paper. The full derivation, numerical comparison, and interpretation are presented in: *NMBS-C7447 at $z = 1.80$ in CPTG: Native Locking and Local Structural Invariance Compared.*

References

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